Euclid's Elements — Book I
Proposition 28

CGJteamLab

If a straight line falling on two straight lines makes the exterior angle equal to the interior and opposite angle on the same side, or the sum of the interior angles on the same side equal to two right angles, then the straight lines are parallel to one another.

1 Introduction

Proposition I.28 gives two sufficient angle conditions for parallelism. A transversal meets two straight lines. Parallelism follows either from equality of a pair of corresponding angles or from the condition that the two interior angles on the same side are together equal to two right angles.

Both branches reduce to Proposition I.27. In each case the essential target is the same equality of alternate interior angles.

No form of the Fifth Postulate is used. Proposition I.28 belongs to neutral geometry.

2 Classical Configuration

Let the straight line EF meet AB at G and CD at H. The configuration is

$$A - G - B, \qquad C - H - D, \qquad E - G - H - F.$$

The proposition has two alternative hypotheses.

2.1 Corresponding-angle criterion

Assume

$$\angle EGB \cong \angle GHD.$$

Since $A - G - B$ and $E - G - H$, the angles

$$\angle EGB \qquad \text{and} \qquad \angle AGH$$

are vertical angles. By Proposition I.15,

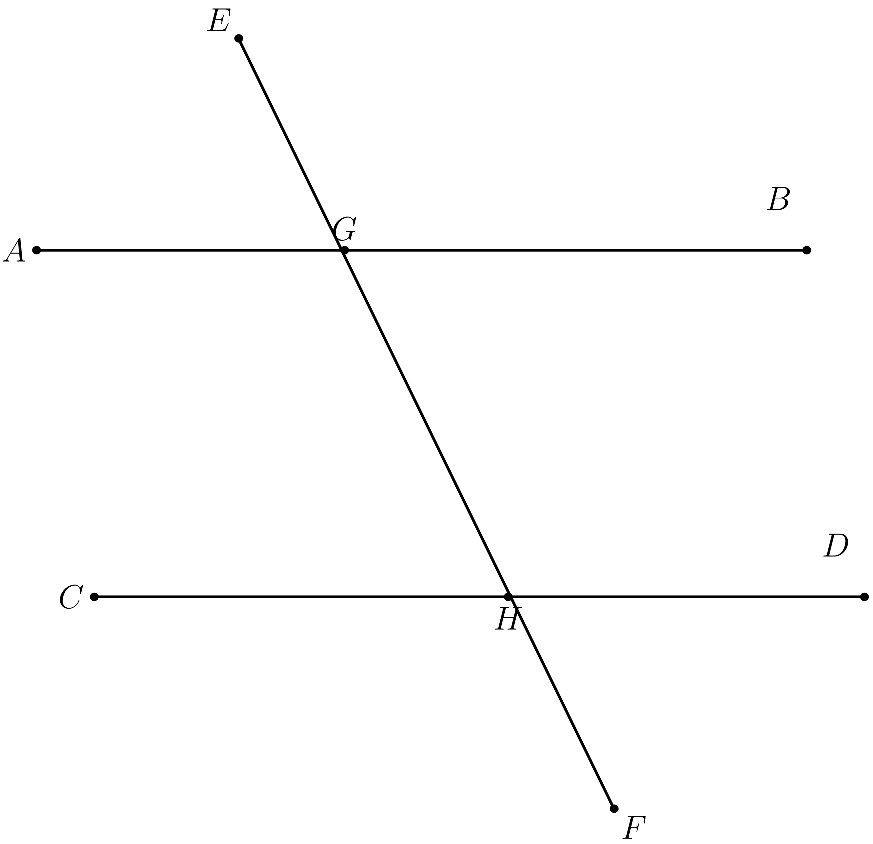

Figure 1: The common transversal configuration for Proposition I.28.

$$\angle AGH \cong \angle EGB.$$

Hence

$$\angle AGH \cong \angle GHD.$$

These are alternate interior angles, so Proposition I.27 gives

$$AB \parallel CD.$$

2.2 Same-side interior-angle criterion

Assume that

$$\angle BGH \quad \text{and} \quad \angle GHD$$

are together equal to two right angles.

Because $A - G - B$, the angles $\angle AGH$ and $\angle BGH$ form a supplementary pair. Proposition I.13 gives the second equality to two right angles. Subtracting the common angle $\angle BGH$ leaves

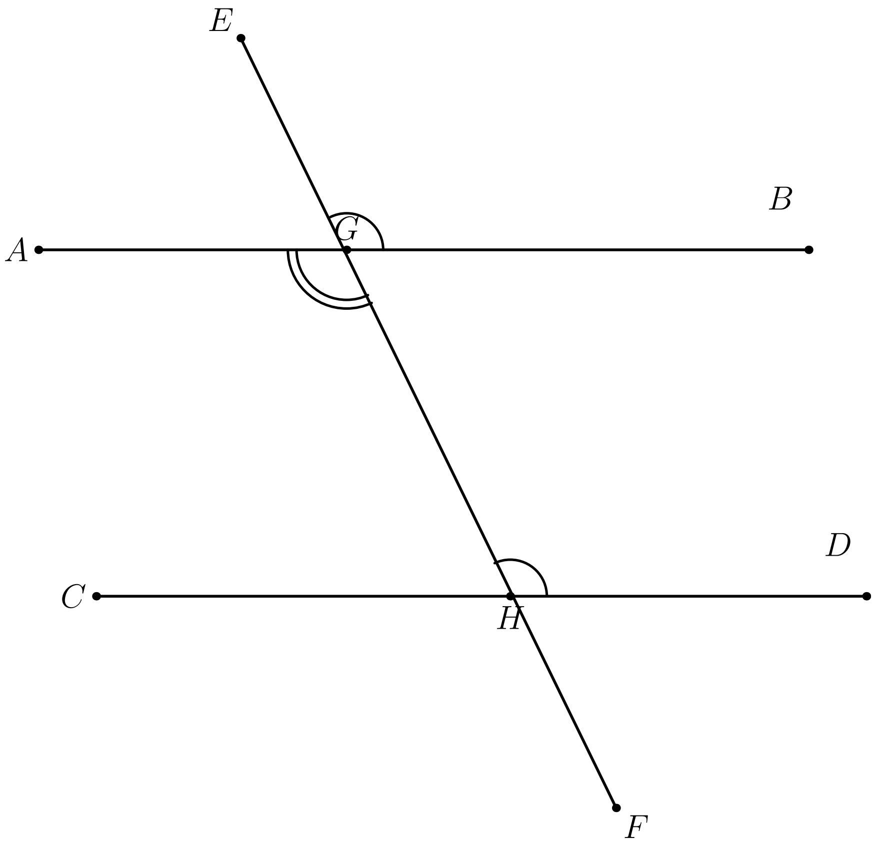

Figure 2: The first criterion. The corresponding angles $\angle EGB$ and $\angle GHD$ are congruent. Proposition I.15 identifies $\angle EGB$ with the vertical angle $\angle AGH$, reducing the argument to Proposition I.27.

$$\angle AGH \cong \angle GHD.$$

Again these are alternate interior angles, and Proposition I.27 gives

$$AB \parallel CD.$$

3 Hilbert Reconstruction

The formal reconstruction preserves the common geometric core of the two branches:

$$\angle AGH \cong \angle GHD \quad \Longrightarrow \quad AB \parallel CD.$$

This final implication is exactly the explicit transversal form of Proposition I.27.

3.1 First branch

The Lean theorem for the corresponding-angle criterion first applies `VerticalAngles`. From the two betweenness relations at G it obtains the vertical-angle congruence. A reversal of the second angle and transitivity with the given corresponding-angle congruence yield

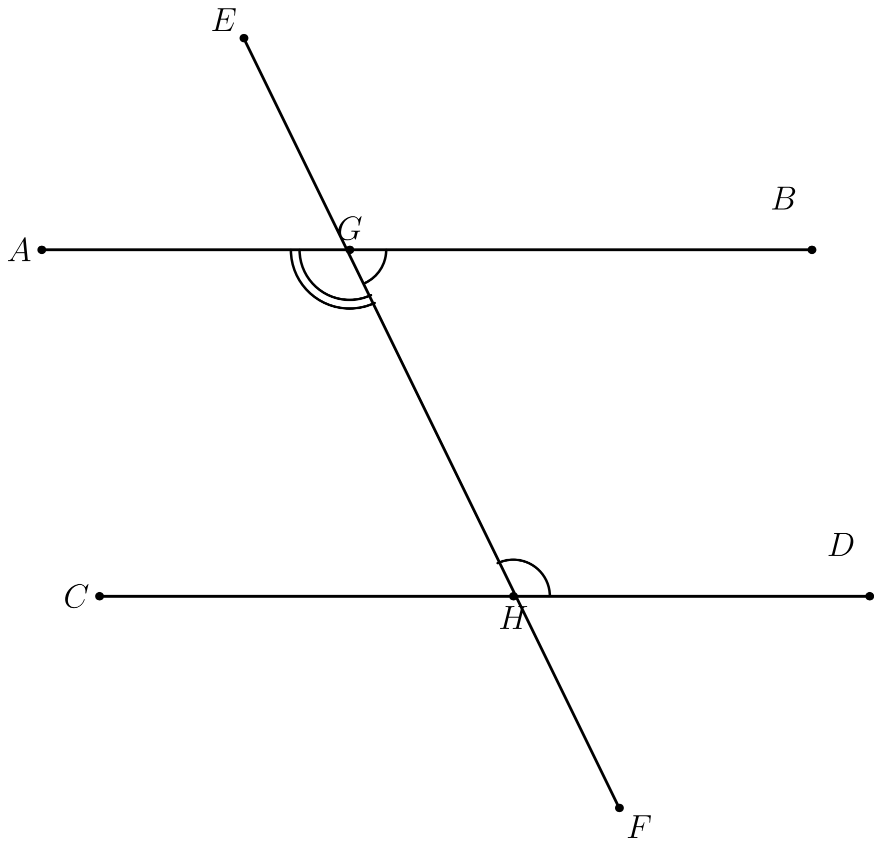

Figure 3: The second criterion. The interior angles $\angle BGH$ and $\angle GHD$ on the same side are together equal to two right angles. The supplementary angle to $\angle BGH$ is $\angle AGH$, so the condition reduces to the alternate-angle hypothesis of Proposition I.27.

$$\angle AGH \cong \angle GHD.$$

The proof then invokes

```
euclid_proposition_27_transversal
```

with G and H as the two intersection points of the transversal.

Schematically,

$$\begin{gathered}
\angle EGB \cong \angle GHD \\
\angle AGH \cong \angle EGB \quad \text{(vertical angles)} \\
\downarrow \\
\angle AGH \cong \angle GHD \\
\downarrow \\
\texttt{euclid_proposition_27_transversal} \\
\downarrow \\
AB \parallel CD.
\end{gathered}$$

3.2 Second branch

The project does not introduce numerical angle measure or an operation of angle addition. The statement that two angles are together equal to two right angles is represented synthetically by an explicit supplementary ray.

The local predicate used in Proposition I.28 is

```
def HilbertAnglesEqualTwoRightAnglesWithSupplement
    (A O B C P D E : Geo.Point) : Prop :=
  Geo.Between D P E /\
  Geo.AngleCongruent A O B C P E
```

For the I.28 configuration it is instantiated as

```
HilbertAnglesEqualTwoRightAnglesWithSupplement
  Geo G H D H G B A
```

and therefore supplies

$$B - G - A$$

together with

$$\angle GHD \cong \angle HGA.$$

Betweenness symmetry recovers $A - G - B$. Symmetry and reversal of angle congruence then give

$$\angle AGH \cong \angle GHD.$$

The proof again ends with

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euclid_proposition_27_transversal
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and hence with $AB \parallel CD$.

4 Neutral Geometry

The logical direction in both parts of Proposition I.28 is

$$\text{angle condition} \quad \Longrightarrow \quad \text{parallel lines.}$$

This direction does not require Hilbert's Group IV. The proof ultimately reduces to the neutral alternate-angle criterion already formalized in Proposition I.27.

Thus Propositions I.27 and I.28 belong to the neutral part of the theory. The reverse direction, from parallelism to the corresponding angle relations, appears in Proposition I.29 and is the point at which Euclidean parallelism enters.

5 Dependency Summary

The two formal branches are

$$\begin{array}{ccc} \text{corresponding-angle congruence} & \qquad & \text{same-side angles equal to } 2R \\ \downarrow & & \downarrow \\ \text{vertical angles} & & \text{synthetic supplementary encoding} \\ \downarrow & & \downarrow \\ \text{alternate interior angles congruent} & & \text{alternate interior angles congruent} \\ \downarrow & & \downarrow \\ \text{I.27} & & \text{I.27} \\ \downarrow & & \downarrow \\ AB \parallel CD & & AB \parallel CD. \end{array}$$

No new parallel axiom is introduced.

6 Conclusion

Proposition I.28 adds two practical recognition criteria for parallel lines, but it does not introduce a new mechanism of parallelism. Both criteria are converted into the alternate-angle condition of Proposition I.27.

The first conversion uses vertical angles. The second uses the synthetic supplementary-angle language already developed in the reconstruction. Consequently the formal proof remains entirely within Hilbert incidence, order, congruence, and the neutral theory of parallelism.